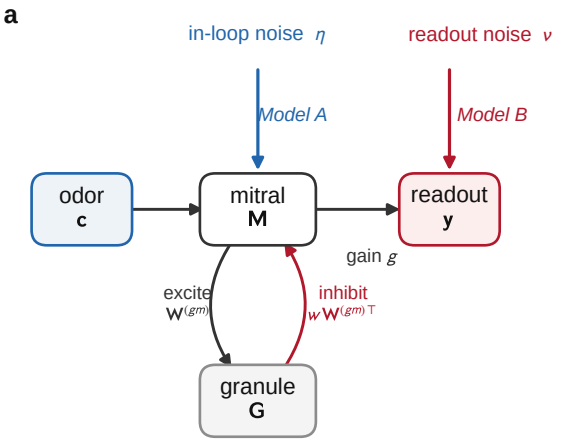

Model A: the loop filters signal and noise alike.
Model B: the loop filters only the signal, and a gain holds total output power fixed.

b

$$\mathcal{I}(\mathbf{c},\mathbf{r}) = \tfrac{1}{2}\log\det[\mathbf{I} + \Sigma_{\eta|c}^{-1}\mathbf{W}^{-1}\mathbf{S}\Sigma_c\mathbf{S}^\top\mathbf{W}^{-\top}]$$

epithelium — *receptor abundances*

$\mathbf{W} = -\mathbf{I},\ \ \mathbf{S} = \mathbf{K}^{1/2}\mathbf{A}$

single neuron — *receptor coexpression*

$N = 1,\ \ M = 2,\ \ \mathbf{S} = (S_1, S_2)$

bulb — *recurrent decorrelation*

$\mathbf{W} = -\mathbf{W}_{\text{eff}},\ \ \mathbf{S} = \mathbf{I}$

feedforward, scalar and full-recurrent limits of one expression